

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Part-III Examination, 2014-15

Paper No. Paper-IX: Statistical Inference & Decision Theory (Old Course)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Question No. 1 is compulsory. Attempt any **four** questions, selecting one from each Unit. Symbols have their usual meanings.*

Question 1 (Compulsory). Answer all parts briefly: [20]

- (a) What is the difference between an estimator and an estimate? [2]
- (b) State the invariance property of a consistent estimator. [2]
- (c) Define a complete statistic. [2]
- (d) When is a statistical test said to be unbiased? [2]
- (e) Define level of significance. [2]
- (f) Define the Minimum Variance Bound (MVB) estimator. [2]
- (g) Define confidence limits and confidence coefficient. [2]
- (h) Define the minimax decision rule. [2]
- (i) Define a statistical parameter with an example. [2]
- (j) Define the power function of a statistical test. [2]

UNIT I

Question 2.

- (a) Define consistency of an estimator. Show that an estimator T_n of parameter θ based on a sample of size n is consistent if:

$$E(T_n) \rightarrow \theta \quad \text{and} \quad \text{Var}(T_n) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

[7]

- (b) What are the desirable properties of a good estimator? Explain any one of them in detail with a suitable example. [7]

UNIT II

Question 3.

- (a) Define a sufficient statistic and state the Fisher–Neyman factorization theorem. [7]
 - (b) State and prove the Rao–Blackwell theorem. [7]
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UNIT III

Question 4.

- (a) Explain the method of interval estimation. Obtain the 95% confidence interval for the variance σ^2 of a normal distribution $N(\mu, \sigma^2)$ when μ is unknown. [7]
- (b) Explain the method of maximum likelihood estimation and state the asymptotic properties of maximum likelihood estimators (MLEs). [7]

Question 5.

- (a) Explain the method of moments for estimating parameters. Discuss its merits and demerits compared to the method of maximum likelihood. [7]
 - (b) Define and explain simple and composite hypotheses with appropriate examples. [7]
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UNIT IV

Question 6.

- (a) Explain the following terms in statistical decision theory:
 - (i) Loss function
 - (ii) Risk function
 - (iii) Bayes' decision rule[7]
- (b) State and prove the Neyman–Pearson lemma. Use it to obtain the most powerful test for testing $H_0: \mu = \mu_0$ against $H_1: \mu = \mu_1$ ($\mu_1 > \mu_0$) on the basis of a sample of size n from $N(\mu, \sigma^2)$ when σ^2 is known. [7]

Question 7.

Describe the procedure of the Likelihood Ratio Test (LRT). Obtain the likelihood ratio test for testing $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$ on the basis of a sample of size n from $N(\mu, \sigma^2)$ when σ^2 is unknown. [14]
